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DSA Series by Shradha Ma'am

Series Link : DSA Series by Shradha Ma'am

	Mark as Done		Question	Link	Difficulty	Video SOLUTION	Companies	Pre-requisites
ARRAYS	☐	1	Majority Element	Problem Link	Easy	Solution Link	Amazon Google	
	☐	2	Repeat & missing number	Problem Link	Easy	Solution Link	Amazon	Hashing
	☐	3	Merge 2 sorted array without xtra space	Problem Link	Easy	Solution Link	Amdocs Brocade Goldman Sachs Juniper Networks	Sorting
	☐	4	Single Number	Problem Link	Easy	Solution Link		
	☐	5	Stock Buy & Sell	Problem Link	Easy	Solution Link	Amazon DEShaw Directi Flipkart Goldman Sachs Int	
	☐	6	Kadane's Algorithm	Problem Link	Medium	Solution Link	Microsoft Facebook	
	☐	7	Pow xn	Problem Link	Medium	Solution Link		
	☐	8	Container with most water	Problem Link	Medium	Solution Link	Flipkart Dunzo	
	☐	9	Sort array of 0s, 1s & 2s	Problem Link	Medium	Solution Link	Microsoft Amazon MakeMyTrip	Sorting
	☐	10	3Sum	Problem Link	Medium	Solution Link	Adobe Amazon Microsoft Morgan Stanley Samsung	Hashing
	☐	11	4Sum	Problem Link	Medium	Solution Link		Hashing
	☐	12	Search a 2D matrix	Problem Link	Medium	Solution Link		2D Array
	☐	13	Next permutation	Problem Link	Medium	Solution Link	Adobe GoldmanSachs Uber	Sorting
	☐	14	Merge overlapping intervals	Problem Link	Medium		Google	Sorting
	☐	15	Longest substring without repeating	Problem Link	Medium		MorganStanley Amazon	String
	☐	16	Set matrix zeroes	Problem Link	Medium		Microsoft Amazon	Sets
	☐	17	Word search	Problem Link	Medium		Ola GoldmanSachs Google	Recursion
	☐	18	Product of array except itself	Problem Link	Medium	Solution Link	Amazon DEShaw Intuit MorganStanley Flipkart	
	☐	19	Subarray sum equals k	Problem Link	Medium			Hashing
	☐	20	Find Duplicate	Problem Link	Medium	Solution Link	Amazon D-E-Shaw Flipkart Paytm Qualcomm Zoho	LL Cycles
	☐	21	Count Inversions	Problem Link	Medium / Hard		Adobe Amazon BankBazaar Flipkart Microsoft Mynt	Merge Sort
	☐	22	Trapping Rainwater	Problem Link	Hard	Solution Link	Samsung	
	☐	23	Sliding window maximum	Problem Link	Hard	Solution Link	Amazon Directi Flipkart Microsoft Google	Hashing
	☐	24	Reverse pairs	Problem Link	Hard			Merge Sort
	☐	25	Largest rectangle in a histogram	Problem Link	Hard	Solution Link		
BINARY SEARCH	☐	1	Search in rotated sorted	Problem Link	Medium	Solution Link	Microsoft Google Adobe Amazon D-E-Shaw Flipkart	
	☐	2	Peak index in mountain array	Problem Link	Medium	Solution Link		
	☐	3	Single element in sorted array	Problem Link	Medium	Solution Link		
	☐	4	Allocate Minimum Pages (Book Allocation)	Problem Link	Medium	Solution Link	Google Infosys Codenation Amazon Microsoft	
	☐	5	Painter's Partition	Problem Link	Medium	Solution Link		
	☐	6	Aggressive cows	Problem Link	Medium	Solution Link	Adobe	
	☐	7	Median of 2 Sorted Arrays	Problem Link	Hard		Amazon Samsung Microsoft Google	
STRINGS	☐	1	Valid Palindrome	Problem Link	Easy	Solution Link	Amazon Cisco DEShaw Facebook FactSet MorganSta	
	☐	2	Longest Common Prefix	Problem Link	Easy		Adobe Blinkit Dunzo	
	☐	3	Valid Anagram	Problem Link	Easy		Google Adobe Flipkart Nagarro Media.net Directi	
	☐	4	Reverse Words in String	Problem Link	Medium	Solution Link		
	☐	5	Remove All Occurrences of String	Problem Link	Medium	Solution Link		
	☐	6	Permutation in String	Problem Link	Medium	Solution Link	Adobe GoldmanSachs Uber	
	☐	7	String Compression	Problem Link	Medium	Solution Link		
	☐	8	Group Anagrams	Problem Link	Medium		Samsung Adobe Amazon	
	☐	9	Minimum Window Substring	Problem Link	Hard		Amazon Google MakeMyTrip Streamoid Technologi	
	☐	10	Kmp Algorithm	Problem Link	Hard			
	☐	11	Robin Karp	--concept--	Hard			
RECURSION & BACKTRACKING	☐	1	Combination Sum I	Problem Link	Medium	Solution Link	Adobe Amazon Microsoft oracle, DE shaw ,salesforc	
	☐	2	Combination Sum II	Problem Link	Medium		Adobe Amazon Microsoft goldmann sach oracle	
	☐	3	Palindrome Partitioning	Problem Link	Medium	Solution Link	meta ,adobe, Infosys, Walmart Labs, Amazon Micro	
	☐	4	Knights Tour	Problem Link	Medium		google,amazon , microsoft, oracle, meta,TCS, apple,	
	☐	5	M Coloring	Problem Link	Medium		google , amazon, microsoft, meta , intuit, citadel, go	
	☐	6	Rat in a Maze	Problem Link	Medium	Solution Link	Amazon	
	☐	7	Subsets II	Problem Link	Medium	Solution Link	google, amazon, microsoft, meta , adobe, apple ,TCS	
	☐	8	Merge Sort	Problem Link	Medium		Google, amazon , meta , microsoft	
	☐	9	N Queens	Problem Link	Hard	Solution Link	Microsoft Amazon	
	☐	10	Sudoku Solver	Problem Link	Hard	Solution Link	Amazon, TCS , Apple, Meta ,Infosys , Oracle, Adobe,	
	☐	11	Count Inversions	Problem Link	Hard		Google, amazon , salessforce	
LINKED LIST	☐	1	Reverse a LL	Problem Link	Easy	Solution Link	google, meta , apple , microsoft, amazon , oracle, a	
	☐	2	Middle of LL	Problem Link	Easy	Solution Link	amazon, meta ,google, goldman sachs, qualcomm,i	
	☐	3	Merge 2 Sorted LL	Problem Link	Easy	Solution Link	Accolite Amazon Belzabar Brocade FactSet Flipkart	
	☐	4	Check if LL Is Palindrome or Not	Problem Link	Easy		amazon, meta ,google, goldman sachs, cisco , sams	
	☐	5	Detect Cycle in LL	Problem Link	Easy	Solution Link	google, meta , microsoft, amazon, oracle, paytm, ap	
	☐	6	Remove Cycle in LL	Problem Link	Medium	Solution Link	amazon, meta , apple,	
	☐	7	Flatten LL	Problem Link	Medium	Solution Link	google, amazon, meta ,microsoft , qualcomm, oracle,	
	☐	8	Clone LL with Random Pointers	Problem Link	Medium	Solution Link	meta , amazon, microsoft , google, intel , walmart la	
	☐	9	Add 2 Numbers	Problem Link	Medium		google, amazon, meta , microsoft, oracle, Tejas netwo	
	☐	10	Reverse Linked List 2	Problem Link	Medium		Google , amazon, meta , microsoft, adobe, uber	
	☐	11	LRU Cache	Problem Link	Medium	Solution Link	Nvidia , Google, Meta , Amazon, Microsoft, Oracle, Ad	
	☐	12	Rotate a LL	Problem Link	Medium		Amazon, Google, Meta , Microsoft ,Salesforce ,Infosys	
	☐	13	Reverse Nodes in K Groups	Problem Link	Hard	Solution Link	google, amazon, meta , microsoft, apple, uber, adobe,	
	☐	1	Implement Stack using Queue	Problem Link	Easy	Solution Link	google , meta , amazon, microsoft, adobe, oracle, o	

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	Mark as Done		Question	Link	Difficulty	Video SOLUTION	Companies	Pre-requisites
STACK & QUEUE	☐	2	Implement Queue using Stack	Problem Link	Easy	Solution Link	amazon, google, microsoft, adobe, oracle , netflix , me	
	☐	3	Next Greater Element I	Problem Link	Easy	Solution Link	amazon, meta , swiggy , microsoft , google, apple, m	
	☐	4	Valid Parenthesis	Problem Link	Easy	Solution Link	Google , meta , amazon, linkedin , intuit , visa , IBM	
	☐	5	1st Non Repeating in Stream	Problem Link	Easy	Solution Link	amazon, goldman sach, google, apple, meta , micro	
	☐	6	Reverse 1st K Elements of Queue	Problem Link	Easy		Microsoft , amdocs, amazon	
	☐	7	Time needed to Buy Tickets	Problem Link	Easy		uber, amazon, microsoft, google, ,meta	
	☐	8	Next Greater Element II	Problem Link	Medium	Solution Link	amazon, uber	
	☐	9	Previous Smaller Element	Problem Link	Medium	Solution Link	amazon, meta , microsoft, apple, visa , oracle, intuit	
	☐	10	Celebrity Problem	Problem Link	Medium	Solution Link	linkedin, meta, microsoft, amazon, uber, salesforce,	
	☐	11	Get Min Element from Stack	Problem Link	Medium	Solution Link	Google, meta, amazon, saleforce, intuit , paypal, adob	
	☐	12	Circular Tour / Gas Station	Problem Link	Medium	Solution Link	amazon, microsoft, goldman sachs, intuit ,IBM	
	☐	13	Rotten Oranges	Problem Link	Medium		amazon, google, microsoft, meta, oracle, adobe, cis	
	☐	14	Stock Span	Problem Link	Medium	Solution Link	Amazon , google, Microsoft, Oracle, Samsung , Meta	
	☐	15	Max Area in Histogram	Problem Link	Hard	Solution Link	Adobe, apple, Meta , Amazon, google, microsoft	
BINARY TREES	☐	1	Inorder	Problem Link	Easy	Solution Link	google , amazon , meta , microsoft ,adobe, uber	
	☐	2	Preorder	Problem Link	Easy	Solution Link	salesforce, amazon, microsoft, meta , google ,adobe	
	☐	3	Postorder	Problem Link	Easy	Solution Link	Amazon ,meta , google, apple, adobe	
	☐	4	Symmetric Tree	Problem Link	Easy		Amazon , Google, Oracle, Apple,meta,Uber,Intuit ,A	
	☐	5	Minimum Distance between Nodes	Problem Link	Easy	ynP1nA&list=PLfqMhTWNE	amazon, google, meta , linkedin , adobe, microsoft	
	☐	6	Are 2 Trees Identical or Not	Problem Link	Easy	Solution Link	meta , google , microsoft,	
	☐	7	Morris Inorder Traversal	Problem Link	Easy	Solution Link	Google,amazon, meta ,apple, adobe, TCS, flipkart , ub	
	☐	8	Diameter	Problem Link	Easy	Solution Link	google, amazon, meta , microsoft , apple, adobe, uber	
	☐	9	Check if Tree is Height Balanced	Problem Link	Easy		amazon , meta , amazon , google ,visa , oracle , TCS	
	☐	10	Subtree of Another Tree	Problem Link	Easy	Solution Link	Amazon google microsoft adobe uber apple	
	☐	11	Check if BT Mirror of itself or not	Problem Link	Easy		Amazon, google, meta, uber , morgan stanley , ebay	
	☐	12	Top View of a Tree	Problem Link	Medium	Solution Link	amazon, google, meta, linkedin , apple, adobe,micro	
	☐	13	Bottom View of a Tree	Problem Link	Medium		meta , amazon, google , uber, oracle, flipkart , JP mo	
	☐	14	Level Order	Problem Link	Medium		josh technology , amazon, google, adobe	
	☐	15	Kth Level of Tree	Problem Link	Medium	Solution Link	meta , google, amazon, adobe, microsoft	
	☐	16	LCA	Problem Link	Medium	Solution Link	meta , amazon , google, linkedin , intuit , oracle,ado	
	☐	17	Transform to Sum Tree	Problem Link	Medium	Solution Link	SAP , Amazon, Ebay , Microsoft	
	☐	18	Construct BT from Inorder & Pre order	Problem Link	Medium	Solution Link	Amazon, Microsoft, meta , adobe, uber , VMware ,	
	☐	19	Construct BT from Inorder & Post order	Problem Link	Medium		Google, amazon, microsoft , adobe, bloomberg	
	☐	20	Flatten BT to LL	Problem Link	Medium	Solution Link	Google, meta , amazon ,microsoft, Myntra , adobe, ora	
	☐	21	Max Width of BT	Problem Link	Medium	Solution Link	Amazon , Meta , Uber, Microsoft, Uber , Adobe,Apple	
	☐	22	Zig Zag Traversal of BT	Problem Link	Medium		Amazon , meta, microsoft, google,oracle, Adobe, Wa	
	☐	23	Max Path Sum	Problem Link	Hard		Google, Meta , Amazon , Oracle, Salesforce, Goldma	
	☐	24	Kth Ancestor	Problem Link	Hard		Google , Amazon ,Microsoft	
BST	☐	1	Kth largest in BST	Problem Link	Easy		Meta , Amazon, Google , Adobe, Atlassian , Salesfor	
	☐	2	Sorted Array to Balanced BST	Problem Link	Easy	Solution Link	Amazon , Google, Meta, Apple, Adobe , Airbnb , Sams	
	☐	3	Validate BST	Problem Link	Medium	Solution Link	Amazon, Google, Meta, Salesforce, IBM , Adobe, Cita	
	☐	4	Kth Smallest in BST	Problem Link	Medium	Solution Link	Accolite Amazon Google Uber Microsoft Meta	
	☐	5	LCA in BST	Problem Link	Medium	Solution Link	Meta, Amazon, Linkedin , Bloomberg, Oracle, Samsu	
	☐	6	Populate Next Right Pointers	Problem Link	Medium	Solution Link	Meta , Amazon, Walmart Labs, Flipkart, Saleforce,Adc	
	☐	7	Recover BST	Problem Link	Medium	Solution Link	Microsoft, amazon, google, oracle, adobe	
	☐	8	Construct from Preorder	Problem Link	Medium	Solution Link	Adobe, Meta , Google, Microsoft	
	☐	9	BST Iterator	Problem Link	Medium	Solution Link	Meta , Amazon, Microsoft, Linkedin , Adobe	
	☐	10	Flatten BST to Sorted list	Problem Link	Medium		Meta , microsoft, Amazon, Nvidia	
	☐	11	Inorder Successor	Problem Link	Medium	Solution Link	Amazon , Google, meta , Citadel, Nvidia,Linkedin ,ub	
	☐	12	Inorder Predecessor	Problem Link	Medium	Solution Link	Microsoft , Bloomberg	
	☐	13	Largest BST in BT	Problem Link	Hard	Solution Link	Microsoft , Meta	
	☐	14	Serialize & Deserialize BST	Problem Link	Hard		Amazon, Google, Meta , Oracle, Flipkart , Adobe, Micr	
	☐	15	Merge 2 BSTS	Problem Link	Hard	Solution Link	Meta, Microsoft	
HEAPS	☐	1	Merge K Sorted Arrays	https://leetcode.com/problen	Easy		google, meta , amazon, microsoft, HCL , Cisco, IBM ,	
	☐	2	K most Frequent Elements	https://leetcode.com/problen	Medium		Meta , Amazon, Google , Microsoft , Oracle, Goldma	
	☐	3	Heap Sort	https://leetcode.com/problen	Medium		Adobe , TCS , Amazon, Google, Meta , hive, Infosys ,	
	☐	4	Kth Smallest Element	https://leetcode.com/problen	Medium		Amazon, Meta , apple, phonepe , salesforce	
	☐	5	Median from Stream	https://leetcode.com/problen	Hard		Google, Amazon, Meta, Apple, Microsoft, Pinterest	
	☐	6	Smallest Range in K Sorted List	https://leetcode.com/problen	Hard		Amazon, Phonepe , Meta , Microsoft , Flipkart , DE sh	
TRIE	☐	1	Longest Common Prefix	https://leetcode.com/problen	Easy		amazon, TCS, Visa, Accenture, Infosys , Deloitte , Go	
	☐	2	Word Break Problem	https://leetcode.com/problen	Medium		Amazon, Google , Netflix , Intuit , Walmart Labs , Sa	
	☐	3	Implement a Phone Directory	https://leetcode.com/problen	Medium		Google	
	☐	4	Implement a Trie	https://leetcode.com/problen	Medium		Google, Amazon, Apple, Meta, Uber, Nvidia , Samsu	
	☐	5	Longest String with All Prefix	https://leetcode.com/problen	Medium		Google	
	☐	1	Flood Fill Algorithm	https://leetcode.com/problen	Easy		Samsung , Flipkart ,SAP Labs , Ola , amazon, micros	
	☐	2	BFS	https://leetcode.com/problen	Medium	Solution Link	Samsung , Intuit, Accolite , Amazon	
	☐	3	DFS	https://leetcode.com/problen	Medium	Solution Link	Amazon, Meta , Google, Visa, Adobe, Microsoft, App	
	☐	4	Detect cycle in undirected using BFS	https://leetcode.com/problen	Medium		Amazon, Meta , Google, Visa, Adobe, Microsoft, App	
	☐	5	Detect cycle in undirected using DFS	https://leetcode.com/problen	Medium		Google , Meta , microsoft, Inmobi , Amazon	
	☐	6	Detect cycle in directed using BFS	https://leetcode.com/problen	Medium		Google , Meta , microsoft, Inmobi , Amazon	

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	Mark as Done		Question	Link	Difficulty	Video SOLUTION	Companies	Pre-requisites
GRAPH	☐	7	Detect cycle in directed using DFS	https://leetcode.com/problen	Medium		Google, Amazon, Meta ,Microsoft, Apple, uber, Adol	
	☐	8	Topological Sorting (DFS)	https://leetcode.com/problen	Medium		Flipkart, Morgan Stanley, Accolite, Amazon , Microsc	
	☐	9	Prim's Algorithm (MST)	https://leetcode.com/problen	Medium		Flipkart, Morgan Stanley, Accolite, Amazon , Microsc	
	☐	10	Bellman Ford Algorithm	https://leetcode.com/problen	Medium		Amazon, Google, Meta, Salesforce, Intuit , Linkedin , I	
	☐	11	Floyd Warshall Algo	https://leetcode.com/problen	Medium		flipkart, microsoft	
	☐	12	Kosaraju Strongly Connected Components	https://leetcode.com/problen	Medium		Amazon, Meta , Stripe, Airbnb , Oracle, DE Shaw, Ap	
	☐	13	Check Bi-partite Graph	https://leetcode.com/problen	Medium		Amazon , Adobe, Directi,Uber,DE shaw, Microsoft	
	☐	14	Number of Islands	https://leetcode.com/problen	Medium		Amazon , Adobe, Directi,Uber,DE shaw, Microsoft	
	☐	15	Rotten Oranges	https://leetcode.com/problen	Medium		Sharechat + Directi , Amazon , microsoft	
	☐	16	01 Matrix	https://leetcode.com/problen	Medium		Google , uber, Meta, microsoft, Amazon	
	☐	17	Course Schedule I & II	https://leetcode.com/problen	Medium		Amazon, Google, Zoho, Oracle, Samsung, Meta , int	
	☐	18	Alien Dictionary	https://leetcode.com/problen	Hard		Google, amazon, Linkedin ,Doordash, Flipkart, Adobe	
	☐	19	Cheapest Flights within K Stops	https://leetcode.com/problen	Medium		Google , amazon , meta , adobe, uber	
	☐	20	Clone a Graph	https://leetcode.com/problen	Medium		Amazon , Microoft , Meta , Linkedin , Oracle , Apple	
	☐	21	Most Stones Removed	https://leetcode.com/problen	Medium		Microsoft Flipkart Pinterest Meta Uber Samung	
	☐	22	Number of Provinces	https://leetcode.com/problen	Medium		Paytm	
	☐	23	Number of Ways to Arrive at Destination	https://leetcode.com/problen	Medium		Phonepe , Google, Apple, Microsoft, Amazon	
	☐	24	Topoplogical Sorting (BFS)	https://leetcode.com/problen	Hard		Sprinklr, Google, Amazon, Meta	
	☐	25	Dijkstra's Algorithm	https://leetcode.com/problen	Hard		Google, Amazon , Meta	
	☐	26	Kruskal's Algorithm (MST)	https://leetcode.com/problen	Hard		Airbnb , Citadel, Uber, Google, Meta , Airbnb	
DP	☐	1	Buy & Sell Stocks I	Problem Link	Easy		Amazon D-E-Shaw Directi Flipkart Goldman Sachs In	
	☐	2	0-1 Knapsack	Problem Link	Medium		Amazon Directi Flipkart GreyOrange Microsoft Mob	
	☐	3	Target Sum Subset	Problem Link	Medium		Myntra ,Microsoft, Meta , Amazon	
	☐	4	Unbounded Knapsack	Problem Link	Medium		Amazon , Google	
	☐	5	Coin Change	Problem Link	Medium		Adobe, Salesforce, Amazon, Google, Microsoft, Intu	
	☐	6	Longest Common Subsequence	Problem Link	Medium		Amazon, Microsoft , Meta , TCS, Oracle	
	☐	7	Longest Common Substring	Problem Link	Medium		Morgan Stanley , Amazon , Microsoft	
	☐	8	Edit Distance	Problem Link	Medium		Goldman Sachs, Paypal , Salesforce , Atlassian, Intuit	
	☐	9	Longest Increasing Subsequence	Problem Link	Medium		Amazon, Google,Linkedin , Cisco , TCS , Meta , Cisc	
	☐	10	Palindromic Partitioning (MCM)	Problem Link	Medium		Google + Goldman Sachs + Citrix , TCS, Meta, Adobe	
	☐	11	Max Product Subarray	Problem Link	Medium		Amazon, Goldman Sachs , Apple,Adobe, TCS, Uber	
	☐	12	Unique BSTs	Problem Link	Medium		Amazon + Google	
	☐	13	Longest Palindromic Subsequence	Problem Link	Medium		Google , Microsoft , Tower Research Capital, Oracle	
	☐	14	Buy & Sell Stocks II	Problem Link	Medium		Walmart + Flipkart	
	☐	15	Nth Catalan	Problem Link	Medium		Amazon , Google, Meta , Microsoft , Infosys	
	☐	16	Minimum Partitioning	Problem Link	Hard		Oracle, Phonepe, Adobe , Apple	
	☐	17	Wildcard Pattern Matching	Problem Link	Hard		Linkedin , DE Shaw , Adobe, Uber, Meta,Amazon	
	☐	18	Rod Cutting	Problem Link	Hard		Uber, samsung, Arcesium , Google, amazon, meta	
	☐	19	Egg Dropping	Problem Link	Hard		Microsoft + Amazon + Ola , Salesforce, Apple	
	☐	20	Longest Bitonic Subsequence	Problem Link	Hard		Amazon D-E-Shaw Goldman Sachs Google Hike Ma	
	☐	21	MCM	Problem Link	Hard		Amazon, microsoft, google, meta	
GREEDY	☐	1	Assign Cookies	Problem Link	Easy		Amazon , Accenture , Uber , Adobe, Meta , Google	
	☐	2	Indian Coins	Problem Link	Medium		Accolite Amazon Morgan Stanley Oracle Paytm San	
	☐	3	Fractional Knapsack	Problem Link	Medium		Microsoft	
	☐	4	maximum length of pair chain	Problem Link	Medium		Amazon , bloomberg, swiggy , adobe , apple	
	☐	5	Activity Selection	Problem Link	Medium		Meta, amazon , google , microsoft	
	☐	6	Job Scheduling	Problem Link	Hard		amazon , microsoft , airbnb, adobe,phonepe	

DSA MASTER CHECKLIST

PLACEMENTS EDITION • JAVA TRACK

How to use this combined PDF

Part 1 contains your original 3-page DSA for Placements sheet, kept unchanged with its existing Problem Links and Video Solution Links. Part 2 adds missing patterns, advanced topics, and extra clickable practice resources.

Recommended order

1. Learn the concept and Java implementation. 2. Solve the original sheet questions. 3. Complete the additional pattern problems. 4. Re-solve mistakes without looking at solutions. 5. Practise timed mixed sets.

Difficulty guidance

Start with fundamentals and Easy problems, move to Medium patterns, then selected Hard problems. Do not rush into advanced data structures before mastering arrays, recursion, binary search, trees, graphs, and core DP.

Note: This is a broad placement-preparation checklist, not a guarantee of covering every question asked by every company. The added links are external learning/practice resources and may change over time.

PART 2 — ADDITIONAL TOPICS & PRACTICE

The original sheet is preserved in Part 1. Use this section to fill topic and pattern gaps.

0. DSA Foundations & Complexity

Before starting problem patterns

Time and Space Complexity

Big-O, Omega, Theta; best/average/worst case; auxiliary space

- Big-O notation basics
- Complexity analysis practice
- Recursion tree method

Java Collections for DSA

ArrayList, LinkedList, HashMap, HashSet, TreeMap, TreeSet, PriorityQueue, ArrayDeque

- Java Collections Framework
- PriorityQueue

Recursion fundamentals

Call stack, base case, recursive tree, stack overflow, backtracking template

- Recursion practice

1. Arrays, Prefix Sum & Hashing Patterns

Build on the Arrays section already present in the original PDF

Prefix Sum

Range sum, 2D prefix sum, difference array

- Range Sum Query - Immutable
- Subarray Sum Equals K
- Corporate Flight Bookings

Two Pointers

Opposite direction, same direction, fast-slow pointers

- Two Sum II
- Remove Duplicates from Sorted Array
- 3Sum

Sliding Window

Fixed window, variable window, frequency window

- Maximum Average Subarray I
- Longest Substring Without Repeating Characters
- Minimum Window Substring
- Permutation in String

HashMap / HashSet patterns

Frequency counting, grouping, lookup, prefix-sum + hashmap

- Group Anagrams
- Longest Consecutive Sequence
- Top K Frequent Elements

Matrix patterns

Spiral traversal, boundary traversal, matrix search, in-place transforms

- Spiral Matrix
- Rotate Image
- Set Matrix Zeroes

2. Binary Search — Advanced Patterns

Beyond the seven binary-search problems in the original PDF

Binary Search on Answer

Monotonic feasibility predicate; minimum feasible / maximum feasible answer

- Koko Eating Bananas
- Capacity To Ship Packages Within D Days
- Split Array Largest Sum

Search Space & Boundaries

Lower bound, upper bound, first/last occurrence, insertion position

- Find First and Last Position
- Search Insert Position

2D Binary Search

Row/column sorted matrices and answer-space search

- Search a 2D Matrix II

3. Bit Manipulation

A major missing topic; learn bit operations before advanced problems

Bitwise fundamentals

AND, OR, XOR, NOT, left/right shift, masks, set/clear/toggle/check bit

- Single Number
- Number of 1 Bits
- Counting Bits

Bitmask techniques

Subsets using masks, XOR properties, bitmask DP introduction

- Subsets
- Maximum XOR of Two Numbers in an Array
- Sum of Two Integers

4. Stack, Queue & Monotonic Structures

Extend the Stack & Queue section in the original PDF

Monotonic Stack

Next greater/smaller, previous greater/smaller, contribution technique

- Daily Temperatures
- Online Stock Span
- Sum of Subarray Minimums

Monotonic Queue

Maintain window maximum/minimum in linear time

- Sliding Window Maximum

Stack Design

Min stack, expression evaluation, decode string

- Min Stack
- Evaluate Reverse Polish Notation
- Decode String

5. Linked List — Additional Patterns

The original sheet already contains core linked-list questions

Fast & Slow Pointer

Cycle, middle, intersection, palindrome

- Linked List Cycle II
- Intersection of Two Linked Lists

Advanced Linked List Design

Doubly linked list, LFU cache, insertion/deletion edge cases

- LFU Cache

6. Trees & BST — Advanced

Extend Binary Trees and BST in the original PDF

Tree Construction & Serialization

Serialize/deserialize binary tree, iterative traversals

- Serialize and Deserialize Binary Tree
- Binary Tree Right Side View

Path & Ancestor Patterns

Path sum, root-to-leaf paths, binary lifting concept

- Path Sum II
- Lowest Common Ancestor of a Binary Tree

Advanced BST

Range queries, predecessor/successor, balanced BST concepts

- Balance a Binary Search Tree

Tree DP / Diameter Patterns

Postorder DP, rerooting idea, subtree aggregation

- House Robber III

7. Heap / Priority Queue — Advanced

Extend the Heap section in the original PDF

Two Heaps Pattern

Median maintenance, partitioning values into lower/upper halves

- Find Median from Data Stream

K-way Merge

Merge multiple sorted lists/arrays using a min heap

- Merge k Sorted Lists
- Find K Pairs with Smallest Sums

Scheduling with Heaps

Events, CPU scheduling, resource allocation

- Meeting Rooms III

8. Graphs — Important Missing Algorithms

Extend the Graph section in the original PDF

Disjoint Set Union (DSU / Union-Find)

Find, union, path compression, union by rank/size

- Redundant Connection
- Accounts Merge
- Number of Provinces

Advanced Shortest Paths

Dijkstra, Bellman-Ford, Floyd-Warshall; 0-1 BFS concept

- Network Delay Time
- Path With Minimum Effort
- Shortest Path in Binary Matrix

MST & Connectivity

Prim, Kruskal, DSU, connected components

- Min Cost to Connect All Points

Bridges & Articulation Points

Tarjan discovery/low-link times; critical connections

- Critical Connections in a Network

Graph Coloring & State

Bipartite graph, multi-source BFS, graph state modeling

- Is Graph Bipartite?
- As Far from Land as Possible

9. Dynamic Programming — Pattern Mastery

Extend the DP section in the original PDF

1D DP

Take/not-take, house robber, climbing stairs, state transitions

- House Robber
- House Robber II
- Decode Ways

Grid DP

Unique paths, obstacles, minimum path sum, falling path

- Unique Paths
- Minimum Path Sum

Knapsack Variants

0/1, unbounded, subset sum, partition, coin change

- Partition Equal Subset Sum
- Coin Change II

LIS Family

LIS O(n log n), number of LIS, envelope nesting

- Number of Longest Increasing Subsequence
- Russian Doll Envelopes

String DP

LCS family, edit distance, palindrome subsequence, wildcard matching

- Distinct Subsequences
- Interleaving String

Interval DP / MCM

Partition intervals, burst balloons, matrix-chain style transitions

- Burst Balloons
- Minimum Cost to Cut a Stick

Tree DP / Bitmask DP

DP on trees and subsets; learn after core DP

- Maximum Students Taking Exam

10. Greedy & Intervals

Extend the Greedy section in the original PDF

Interval Scheduling

Merge, insert, erase overlaps, meeting rooms

- Merge Intervals
- Insert Interval
- Non-overlapping Intervals

Greedy Proof & Scheduling

Sorting criteria, exchange argument, deadline scheduling

- Jump Game
- Jump Game II
- Gas Station

11. Trie & String Algorithms

Extend Trie and String sections in the original PDF

KMP / Prefix Function

Pattern matching in linear time

- Find the Index of the First Occurrence

Z Algorithm & Rolling Hash

Prefix matching, hash-based string search; understand collisions

- Repeated String Match
- Longest Duplicate Substring

Advanced Trie

Word search, prefix count, XOR trie

- Word Search II

12. Range Query Data Structures

Advanced topics useful for stronger product-company preparation

Segment Tree

Build, point update, range query; lazy propagation basics

- Segment Tree introduction

Fenwick Tree / Binary Indexed Tree

Prefix sums with updates; coordinate compression

- Fenwick Tree introduction

Sparse Table

Static range minimum/maximum queries

- Sparse Table introduction

13. Placement Readiness & Revision

Use this after completing topic-wise learning

Core interview patterns

Revise arrays, hashing, two pointers, sliding window, binary search, trees, graphs, DP

- LeetCode Top Interview 150
- LeetCode Interview Crash Course

Timed practice

Solve unseen problems under time limits; explain approach and complexity aloud

- LeetCode problem set
- GeeksforGeeks practice

Revision system

Maintain an error log; revisit solved problems after 1 day, 1 week, and 1 month

- NeetCode Roadmap

JAVA-SPECIFIC IMPLEMENTATION CHECKLIST

■ Arrays and strings: indexing, char arrays, StringBuilder, sorting, comparator.

■ Collections: ArrayList, LinkedList, HashMap, HashSet, TreeMap, TreeSet, PriorityQueue, ArrayDeque.

■ Custom ordering: Comparable, Comparator, sorting objects and pairs.

■ Graphs: adjacency list with List>, weighted edges, visited arrays, BFS queue, DFS recursion/stack.

■ Heap: PriorityQueue min-heap by default; reverse comparator for max-heap.

■ DP: define state, transition, base case, iteration order; memoization before tabulation; space optimization.

■ Recursion: understand call stack and avoid shared mutable-state bugs.

■ Use long when sums/products may exceed int; check integer overflow.

■ Know Java equals/hashCode contract when using custom objects as HashMap/HashSet keys.

■ Write clean helper methods and test empty, singleton, duplicate, sorted, reverse-sorted, and boundary cases.

WEEKLY REVISION CHECKLIST

Day	Focus
Day 1	Arrays, hashing, prefix sum, two pointers
Day 2	Sliding window, strings, binary search
Day 3	Linked list, stack, queue, monotonic stack
Day 4	Trees, BST, heap, trie
Day 5	Graphs, BFS/DFS, DSU, shortest paths, MST
Day 6	DP patterns, greedy, intervals
Day 7	Timed mixed practice + revise mistakes

Progress rule: mark a question complete only when you can explain the approach, state time/space complexity, code it independently in Java, and solve it again later without hints.